

GIRRAWEE HIGH SCHOOL

YEAR 11 MATHEMATICS EXTENSION 2 TASK 1

Time allowed: 110 minutes

December 2010

Instructions:

- Show all necessary working.
- Marks may be deducted for careless or badly arranged work.
- Approved scientific calculators may be used.

Question 1 (12 marks)

(a) Evaluate:

4

(i) i^{25}

(ii) i^{-37}

(iii) $\sum_{r=0}^{1000} (i)^r$

(b) If $z_1 = -2 + 3i$ and $z_2 = 3 - 4i$, express the following in the form $a + ib$.

8

(i) $z_1 + z_2$

(ii) $z_2 - z_1$

(iii) $\frac{z_1 z_2}{z_2}$

Question 2 (13 marks)

(a) Express in modulus argument form where argument is in radians.

6

(i) $z = -1 - i$

(ii) $z = -3 + i\sqrt{3}$

(b) Evaluate each of the following in Cartesian form

(i) $[3(\cos 40^\circ + i \sin 40^\circ)][4(\cos 80^\circ + i \sin 80^\circ)]$

3

(ii) $\frac{(2cis15^\circ)^7}{(4cis45^\circ)^3}$

4

Question 3 (12 marks)

- (a) (i) Find all real numbers x and y such that $(x + iy)^2 = -5 - 12i$. 4
- (ii) Hence solve the equation $z^2 - 4z + (9 + 12i) = 0$ 4
- (b) Solve the equation $z^4 = 8(\sqrt{3} + i)$. Write answers in modulus argument form. 4

Question 4 (27 marks)

- (a) By letting $z = \cos \theta + i \sin \theta$ and expanding z^6 find expressions for
- (i) $\sin 6\theta$ 5
- (ii) $\cos 6\theta$ 4
- (iii) $\tan 6\theta$ in terms of $\tan \theta$ 4
- (b) Prove that
- (i) $z^n + \frac{1}{z^n} = 2\cos n\theta$ (ii) $z^n - \frac{1}{z^n} = 2i \sin n\theta$ 4
- (c) Using the result/s of (b)
- (i) Prove that $\sin^6 \theta = \frac{-1}{32}(\cos 6\theta - 6\cos 4\theta + 15\cos 2\theta - 10)$ 5
- (ii) Find all complex numbers z such that $\frac{(z^2 - \frac{1}{z^2}) \times i}{(z^2 + \frac{1}{z^2})} = -\sqrt{3}$. 5

Question 5 (17 marks)

- (i) Find the seventh roots of unity. Show the roots on an Argand diagram. 6
- (ii) Resolve $z^7 - 1$ into real quadratic and real linear factors. 4
- (iii) Find the sum of the roots. 2
- (iv) Using (iii) above and the fact that $\cos(360 - \theta) = \cos \theta$, deduce that
- $$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$$
- 5

Question 6 (17 marks)(a) If ω is a complex cube root of unity

(i) Show that $(1 - \omega - \omega^2)(1 - \omega + \omega^2)(1 + \omega - \omega^2) = 8$ 4

(ii) Form a quadratic equation with roots $a\omega + b\omega^2$, $a\omega^2 + b\omega$. 4

(b) If $x + iy = \frac{a + ib}{c + id}$, prove that $x^2 + y^2 = \frac{a^2 + b^2}{c^2 + d^2}$ 4

(c) Prove that $\left(\frac{1 + \cos \theta + i \sin \theta}{1 + \cos \theta - i \sin \theta} \right)^n = \cos n\theta + i \sin n\theta$ 5

(Hint: Express $1 + \cos \theta$ and $\sin \theta$ in terms of $\frac{\theta}{2}$)**Question 7 (26 marks)**(a) If $z = 3 + 2i$, plot on an Argand diagram 5

(i) z (ii) $-z$ (iii) $\bar{z}$

(iv) iz (v) $z + 2i$

(b) Sketch these on separate Argand diagrams.

(i) $|z + 2 + 3i| < 2$ 2

(ii) $1 \leq |z - 1| < 3$ 3

(iii) $\frac{\pi}{6} \leq \arg z < \frac{\pi}{3}$ 3

(iv) $\arg(z - 1 - 2i) = \frac{\pi}{4}$. Find the equation of the Locus. 5

(v) $2 \leq |z| \leq 3$ and $\text{Im}(z) \geq 1$ 3

(c) Prove that for any two complex numbers z and ω

$|z + \omega|^2 + |z - \omega|^2 = 2[|z|^2 + |\omega|^2]$. Give a geometrical interpretation of this equation. 5

Question 8 (14 marks)

- (a) Plot the points $z = i$ and $\omega = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$ on an Argand diagram. Deduce from the diagram that $\tan \frac{3\pi}{8} = \sqrt{2} + 1$. (Hint: $|z| = |\omega| = 1$, $\arg z = \frac{\pi}{2}$, $\arg \omega = \frac{\pi}{4}$) 4

- (b) Find the locus of z :

- (i) If $\arg \frac{z-3}{z-1} = \frac{\pi}{4}$ (Find the centre and radius of the corresponding circle.

Show working and write reasons) 6

- (ii) If $\left| \frac{z-1}{z+2} \right| = 2$ (Find the algebraic equation and describe it) 4

Question 9 (14 marks)

Let $x = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$ and $b = x + \frac{1}{x}$, $b > 0$.

- (i) Show that $x^4 + x^3 + x^2 + x + 1 = 0$. 3

- (ii) Use (i) to show that $b^2 + b - 1 = 0$. 2

- (iii) Show that $x^2 - bx + 1 = 0$ and hence $x = \frac{1}{2}(b + i\sqrt{4-b^2})$ 4

- (iv) Hence show that $\cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4}$ and $\sin \frac{2\pi}{5} = \frac{1}{2} \sqrt{\frac{5 + \sqrt{5}}{2}}$. 5

Extension 2 Task 1, 2010 Solutions

Question 1 (12 marks)

(a) (i) $i^{25} = i^{24} \times i$

$$= (i^4)^6 \times i$$

$$= i$$

(ii) $i^{-37} = \frac{1}{i^{37}}$

$$= \frac{1}{i^{36} \times i}$$

$$= \frac{1}{i^4 \times i}$$

$$= \frac{1}{i^5}$$

$$= \frac{1}{i} = \frac{i}{i^2}$$

$$= -i$$

$$= -i$$

$$= -i$$

(iii) $\sum_{n=0}^{1000} (i)^n$

$$n=0$$

$$= i^0 + i^1 + i^2 + i^3 + \dots + i^{997} + i^{998} + i^{999} + i^{1000}$$

$$= 1 + (i - 1 - i + 1) + (i - 1 - i + 1) + \dots + (i - 1 - i + 1)$$

$$= 1$$

$$= 1$$

$$= 1$$

(b) $z_1 = -2 + 3i$ $z_2 = 3 - 4i$

(i) $z_1 + z_2 = -2 + 3i + 3 - 4i$

$$= (-2 + 3) + i(3 - 4)$$

$$= 1 - i$$

$$= 1 - i$$

(ii) $z_2 - z_1 = 3 - 4i - (-2 + 3i)$

$$= 3 - 4i + 2 - 3i = 5 - 7i$$

$$= 5 - 7i$$

Question 2 (13 marks)

(a) (i) $z = -1 - i$

$$r = \sqrt{1+1} = \sqrt{2}$$

$$\tan \alpha = \frac{1}{1} = 1$$

$$\alpha = \frac{\pi}{4}$$

$$\theta = -\pi + \alpha$$

$$= -\pi + \frac{\pi}{4}$$

$$= -\frac{4\pi + \pi}{4}$$

$$= -\frac{5\pi}{4}$$

$$= -\frac{5\pi}{4}$$

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$$= -\frac{5\pi}{4}$$

(ii) $(2 \operatorname{cis} 15^\circ)^7 = \frac{2^7 \operatorname{cis} 105^\circ}{(4 \operatorname{cis} 45^\circ)^3}$

$$= \frac{2^7 \operatorname{cis} 105^\circ}{64 \operatorname{cis} 135^\circ}$$

$$= 2 \operatorname{cis} (105 - 135)$$

$$= 2 \operatorname{cis} (-30)$$

$$= 2 (\cos(-30) + i \sin(-30))$$

$$= 2 (\cos 30 - i \sin 30)$$

$$= 2 \left(\frac{\sqrt{3}}{2} - i \times \frac{1}{2} \right) = \sqrt{3} - i$$

$$= \sqrt{3} - i$$

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Question 3 (12 marks)

(a) (i) $(x+iy)^2 = -5-12i$

$$x^2 + 2ixy - y^2 = -5-12i$$

$$x^2 - y^2 = -5 \quad \text{--- (1)}$$

$$2xy = -12$$

$$xy = -6 \quad \text{--- (2)}$$

From (2) $y = \frac{-6}{x}$

Substitute in (1)

$$x^2 - \left(\frac{-6}{x}\right)^2 = -5 \quad \text{--- (A)}$$

$$x^2 - \frac{36}{x^2} = -5$$

$$x^4 - 36 = -5x^2$$

$$x^4 + 5x^2 - 36 = 0$$

$$(x^2 + 9)(x^2 - 4) = 0$$

$$x^2 = -9 \quad \text{or} \quad x^2 = 4$$

$$x = \pm 2$$

When $x = 2$, $y = \frac{-6}{2} = -3$

When $x = -2$, $y = \frac{-6}{-2} = 3$

The square roots of

$-5-12i$ are $2-3i$ and

$-2+3i$
 $\pm(2-3i)$

(ii) $z^2 - 4z + (9+12i) = 0$ Page 3

$$z = \frac{4 \pm \sqrt{16 - 4 \times 1 \times (9+12i)}}{2}$$

$$= \frac{4 \pm \sqrt{16 - 36 - 48i}}{2}$$

$$= \frac{4 \pm \sqrt{-20 - 48i}}{2} = \frac{4 \pm \sqrt{4(-5-12i)}}{2}$$

$$= \frac{4 \pm 2\sqrt{-5-12i}}{2}$$

$$= \frac{4 \pm 2(2-3i)}{2} \quad \text{--- (4)}$$

$$= \frac{4 + 2(2-3i)}{2} \quad \text{or} \quad \frac{4 - 2(2-3i)}{2}$$

$$= \frac{4 + 4 - 6i}{2} \quad \text{or} \quad \frac{4 - 4 + 6i}{2}$$

$$= \frac{8-6i}{2} \quad \text{or} \quad \frac{6i}{2}$$

$$= 4-3i \quad \text{or} \quad 3i$$

(b) $z^4 = 8(\sqrt{3}+i)$ $z = [8(\sqrt{3}+i)]^{\frac{1}{4}}$

$$\frac{\sqrt{3}+i}{2} \quad r = \sqrt{3+1} = 2$$

$$\tan \alpha = \frac{1}{\sqrt{3}} \quad \alpha = \frac{\pi}{6}$$

$$\theta = \frac{\pi}{6}$$

$$\sqrt{3}+i = 2 \operatorname{cis} \frac{\pi}{6}$$

$$8(\sqrt{3}+i) = 16 \operatorname{cis} \frac{\pi}{6}$$

where k is an integer

$$= 16 \operatorname{cis} \frac{12k\pi + \pi}{6}$$

$$z = [16 \operatorname{cis} \frac{12k\pi + \pi}{6}]^{\frac{1}{4}}$$

$$= 2 \operatorname{cis} \frac{12k\pi + \pi}{24}$$

$$\frac{k=0}{z_1} = 2 \operatorname{cis} \frac{\pi}{24}$$

$$k=1$$

$$z_2 = 2 \operatorname{cis} \frac{13\pi}{24} \quad \text{--- (A)}$$

$$k=2$$

$$z_3 = 2 \operatorname{cis} \frac{25\pi}{24}$$

$$= 2 \operatorname{cis} \left(2\pi - \frac{25\pi}{24} \right)$$

$$= 2 \operatorname{cis} \left(\frac{48\pi - 25\pi}{24} \right)$$

$$= 2 \operatorname{cis} - \left(\frac{23\pi}{24} \right)$$

$$= 2 \operatorname{cis} - \frac{23\pi}{24}$$

$$k=3$$

$$z_4 = 2 \operatorname{cis} \frac{37\pi}{24}$$

$$= 2 \operatorname{cis} - \left(\frac{2\pi - 37\pi}{24} \right)$$

Question 4 (27 marks)

(a) $z^6 = (\cos \theta + i \sin \theta)^6 = \cos 6\theta + i \sin 6\theta$ by De Moivre's theorem

Using binomial theorem we have

$$\begin{aligned} (\cos \theta + i \sin \theta)^6 &= \cos^6 \theta + 6C_1 \cos^5 \theta (i \sin \theta) \\ &+ 6C_2 \cos^4 \theta (i \sin \theta)^2 + 6C_3 \cos^3 \theta (i \sin \theta)^3 \\ &+ 6C_4 \cos^2 \theta (i \sin \theta)^4 + 6C_5 \cos \theta (i \sin \theta)^5 + (i \sin \theta)^6 \\ &= \cos^6 \theta + 6 \cos^5 \theta \times i \sin \theta + 15 \cos^4 \theta \times i^2 \sin^2 \theta \\ &+ 20 \cos^3 \theta \times i^3 \sin^3 \theta + 15 \cos^2 \theta \times i^4 \sin^4 \theta \end{aligned}$$

$$+ 6 \cos \theta \times i^5 \sin^5 \theta + i^6 \sin^6 \theta$$

$$= \cos^6 \theta + 6 \cos^5 \theta i \sin \theta - 15 \cos^4 \theta \sin^2 \theta$$

$$- i 20 \cos^3 \theta \sin^3 \theta + 15 \cos^2 \theta \sin^4 \theta + i 6 \cos \theta \sin^5 \theta$$

$$+ \sin^6 \theta$$

Equating real and imaginary parts

i) $\sin 6\theta = 6 \cos^5 \theta \sin \theta - 20 \cos^3 \theta \sin^3 \theta + 6 \cos \theta \sin^5 \theta$

ii) $\cos 6\theta = \cos^6 \theta - 15 \cos^4 \theta \sin^2 \theta + 15 \cos^2 \theta \sin^4 \theta - \sin^6 \theta$

(iii) $\tan 6\theta = \frac{\sin 6\theta}{\cos 6\theta}$

$$= \frac{6 \cos^5 \theta \sin \theta - 20 \cos^3 \theta \sin^3 \theta + 6 \cos \theta \sin^5 \theta}{\cos^6 \theta - 15 \cos^4 \theta \sin^2 \theta + 15 \cos^2 \theta \sin^4 \theta - \sin^6 \theta}$$

Divide both numerator and denominator by $\cos^6 \theta$ we get

$$= \frac{6 \tan \theta - 20 \tan^3 \theta + 6 \tan^5 \theta}{1 - 15 \tan^2 \theta + 15 \tan^4 \theta - \tan^6 \theta} \quad (4)$$

(b) (i) $z = \cos \theta + i \sin \theta$

$$z^n = (\cos \theta + i \sin \theta)^n$$

$$= \cos n\theta + i \sin n\theta$$

$$\frac{1}{z^n} = z^{-n} = (\cos \theta + i \sin \theta)^{-n}$$

$$= \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos n\theta - i \sin n\theta$$

$$z^n + \frac{1}{z^n} = \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$$

$$\frac{z^n + \frac{1}{z^n}}{2} = \frac{2 \cos n\theta}{2}$$

$$z^n - \frac{1}{z^n} = \cos n\theta + i \sin n\theta - \cos n\theta + i \sin n\theta$$

$$= \frac{2i \sin n\theta}{2}$$

$$(c) (i) (2i \sin \theta)^6 = \left(z - \frac{1}{z}\right)^6$$

$$= z^6 - 6C_1 z^5 \times \frac{1}{z} + 6C_2 z^4 \left(\frac{1}{z}\right)^2 - 6C_3 z^3 \left(\frac{1}{z}\right)^3 + 6C_4 z^2 \left(\frac{1}{z}\right)^4 - 6C_5 z \left(\frac{1}{z}\right)^5 + \left(\frac{1}{z}\right)^6$$

$$= z^6 - 6z^4 + 15z^2 - 20 + \frac{15}{z^2} - \frac{6}{z^4} + \frac{1}{z^6}$$

$$= \left(z^6 + \frac{1}{z^6}\right) - 6\left(z^4 + \frac{1}{z^4}\right) + 15\left(z^2 + \frac{1}{z^2}\right) - 20$$

$$= 2\cos 6\theta - 6 \times 2\cos 4\theta + 15 \times 2\cos 2\theta - 20$$

$$= 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$$

$$64 \times^{-1} \sin 6\theta = 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$$

$$\sin 6\theta = 2\left(\cos 6\theta - 6\cos 4\theta + 15\cos 2\theta - 10\right)$$

$$-64$$

$$= \frac{-1}{32} (\cos 6\theta - 6\cos 4\theta + 15\cos 2\theta - 10)$$

$$\underline{\underline{32}}$$

$$(ii) z^2 - \frac{1}{z^2} = 2i \sin 2\theta \quad \text{and} \quad z^2 + \frac{1}{z^2} = 2\cos 2\theta$$

$$\left(z^2 - \frac{1}{z^2}\right) \times i = -\sqrt{3} \quad \text{can be written as}$$

$$\left(z^2 + \frac{1}{z^2}\right)$$

$$\frac{(2i \sin 2\theta) \times i}{2\cos 2\theta} = -\sqrt{3}$$

$$\text{ie } \frac{-2 \sin 2\theta}{2\cos 2\theta} = -\sqrt{3}$$

$$\text{from } 2\theta = \sqrt{3}, 0 \leq \theta \leq 360^\circ$$

$$\text{let } u = 2\theta \quad 0 \leq u \leq 720^\circ$$

$$\text{from } u = \sqrt{3}$$

$$u = \frac{\pi}{3}, \frac{\pi}{3} + \pi, \frac{\pi}{3} + 2\pi, \frac{\pi}{3} + 3\pi$$

$$= \frac{\pi}{3}, \frac{4\pi}{3}, \frac{7\pi}{3}, \frac{10\pi}{3} \quad (5)$$

$$\theta = \frac{u}{2} = \frac{\pi}{6}, \frac{4\pi}{6}, \frac{7\pi}{6}, \frac{10\pi}{6}$$

$$= \frac{\pi}{6}, \frac{2\pi}{3}, \frac{7\pi}{6}, \frac{5\pi}{3}$$

$$z = \cos \theta + i \sin \theta$$

$$= \underline{\underline{\cos \frac{\pi}{6}, \cos \frac{2\pi}{3}, \cos \frac{7\pi}{6}, \cos \frac{5\pi}{3}}}$$

Question 5 (17 marks)

$$(i) z^7 = 1 = \cos 2k\pi + i \sin 2k\pi$$

$$z = (\cos 2k\pi + i \sin 2k\pi)^{\frac{1}{7}}$$

$$= \cos \frac{2k\pi}{7} + i \sin \frac{2k\pi}{7}$$

$$k = 0, 1, 2, 3, 4, 5, 6$$

$$(6)$$

$$\frac{k=0}{z_1} = \cos 0 + i \sin 0 = 1$$

$$\frac{k=3}{z_4} = \cos \frac{6\pi}{7} + i \sin \frac{6\pi}{7}$$

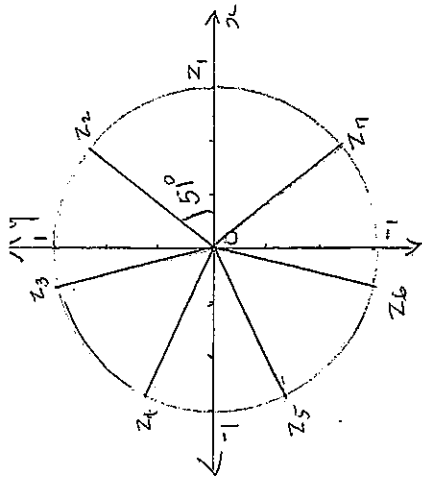
$$\frac{k=1}{z_2} = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$$

$$\frac{k=4}{z_5} = \cos \frac{8\pi}{7} + i \sin \frac{8\pi}{7}$$

$$\frac{k=2}{z_3} = \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7}$$

$$z_6 = \cos \frac{10\pi}{7}$$

$$z_7 = \cos \frac{12\pi}{7}$$



$$\begin{aligned}
 \text{(ii)} \quad z^7 - 1 &= (z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5)(z - z_6)(z - z_7) \\
 &= (z - 1) \prod_{k=1}^6 (z - z_k) \prod_{k=1}^6 (z - \overline{z_k}) \\
 &= (z - 1) [z^2 - z(z_2 + \overline{z_2}) + z_2 \overline{z_2}] [z^2 - z(z_3 + \overline{z_3}) + z_3 \overline{z_3}] \\
 &\quad [z^2 - z(z_4 + \overline{z_4}) + z_4 \overline{z_4}] [z^2 - z(z_5 + \overline{z_5}) + z_5 \overline{z_5}]
 \end{aligned}$$

$$\begin{aligned}
 z_2 + \overline{z_2} &= z_2 + \overline{z_2} & z_3 + \overline{z_3} &= z_3 + \overline{z_3} & z_4 + \overline{z_4} &= z_4 + \overline{z_4} \\
 &= 2 \cos \frac{2\pi}{7} & &= 2 \cos \frac{4\pi}{7} & &= 2 \cos \frac{6\pi}{7} \\
 z_2 z_1 &= z_2 \overline{z_2} & z_3 z_6 &= z_3 \overline{z_3} & z_4 z_5 &= z_4 \overline{z_4} \\
 &= |z_2|^2 = 1 & &= |z_3|^2 = 1 & &= |z_4|^2 = 1
 \end{aligned}$$

$$\begin{aligned}
 z^7 - 1 &= (z - 1) (z^2 - 2 \cos \frac{2\pi}{7} z + 1) (z^2 - 2 \cos \frac{4\pi}{7} z + 1) (z^2 - 2 \cos \frac{6\pi}{7} z + 1) \\
 &\quad (z^2 - 2 \cos \frac{8\pi}{7} z + 1)
 \end{aligned}$$

$$\text{(iii)} \quad z_1 + z_2 + z_3 + z_4 + z_5 + z_6 + z_7 =$$

$$\begin{aligned}
 &= \left(1 + \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{10\pi}{7} + \cos \frac{12\pi}{7} \right) \\
 &\quad + i \left(\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{6\pi}{7} + \sin \frac{8\pi}{7} + \sin \frac{10\pi}{7} + \sin \frac{12\pi}{7} \right) \\
 &= -\frac{\text{coefficient of } z^6}{\text{coefficient of } z^7} = \frac{-0}{1} = 0
 \end{aligned}$$

(iv) A complex number equal to zero means its real part and imaginary part are separately equal to zero

From (iii) we have

$$1 + \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{10\pi}{7} + \cos \frac{12\pi}{7} =$$

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{10\pi}{7} + \cos \frac{12\pi}{7} = -1$$

$$\therefore \cos \frac{8\pi}{7} = \cos \frac{2\pi - 8\pi}{7} = \cos \frac{6\pi}{7}$$

$$\cos \frac{10\pi}{7} = \cos \frac{2\pi - 10\pi}{7} = \cos \frac{4\pi}{7}$$

$$\cos \frac{12\pi}{7} = \cos \frac{2\pi - 12\pi}{7} = \cos \frac{2\pi}{7}$$

(5)

$$2 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) = -1$$

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$$

Question 6 (17 marks)

(a)(i) $1+w+w^2 = 0$

$1-w-w^2 = 1-(w+w^2)$

$= 1-(-1) = 2$

$1-w+w^2 = -w-w^2-w+w^2$
 $= -2w$

$1+w-w^2 = -w^2-w^2$
 $= -2w^2$ (A)

$(1-w-w^2)(1-w+w^2)(1+w-w^2)$
 $= 2 \times -2w \times -2w^2$

$= 8w^3 = 8$

(ii) Sum of roots

$= aw+bw^2+aw^2+bw$

$= a(w+w^2)+b(w+w^2)$

$= (w+w^2)(a+b)$

$= -(a+b)$ (A)

Product of roots

$= (aw+bw^2)(aw^2+bw)$

$= a^2w^3+abw^2+abw^4+b^2w^3$

$= a^2 + ab(w^2+w^4)+b^2$

$= a^2+ab(w^2+w)+b^2$

a^2-ab+b^2

The quadratic equation whose roots are $aw+bw^2$

and aw^2+bw is given by

$z^2 - (a+b)z + (a^2-ab+b^2) = 0$

(b) $z+iy = \frac{a+ib}{c+id}$ — (1)

Taking conjugates we get

$\frac{z+iy}{c+id} = \frac{a+ib}{c+id} = \frac{a+ib}{c+id}$

$z-iy = \frac{a-ib}{c-id}$ — (2)

Multiply (1) and (2)

$(z+iy)(z-iy) = \frac{a+ib}{c+id} \times \frac{a-ib}{c-id}$

$z^2-(iy)^2 = \frac{a^2-(ib)^2}{c^2-(id)^2}$ (A)

$z^2+y^2 = \frac{a^2+b^2}{c^2+d^2}$

(c) LHS = $\left(\frac{1+\cos\theta + i\sin\theta}{1+\cos\theta - i\sin\theta} \right)^n$

$= \left[\frac{2\cos^2\frac{\theta}{2} + i2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2} - i2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} \right]^n$

$= \left[\frac{2\cos\frac{\theta}{2}(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2})}{2\cos\frac{\theta}{2}(\cos\frac{\theta}{2} - i\sin\frac{\theta}{2})} \right]^n$

$= \left[\frac{\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}}{\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}} \right]^n$ (5)

$= \left[\cos\frac{\theta}{2} + i\sin\frac{\theta}{2} \right]^n \left[\cos\frac{\theta}{2} - i\sin\frac{\theta}{2} \right]^{-n}$

$= \left[\cos\frac{\theta}{2} + i\sin\frac{\theta}{2} \right]^n \left[(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2})^{-1} \right]^{-n}$

$= \left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2} \right)^n \left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2} \right)^n$

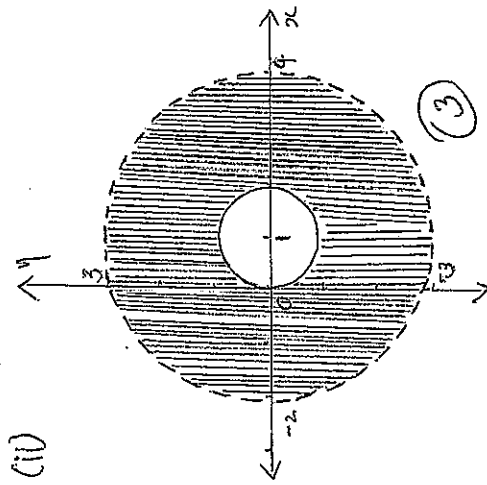
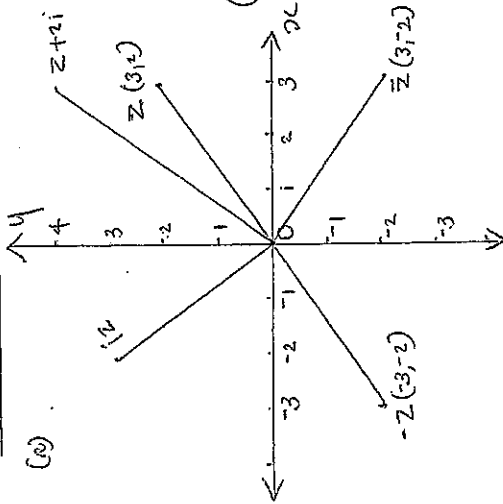
$= \left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2} \right)^{2n}$

$= \cos\frac{2n\theta}{2} + i\sin\frac{2n\theta}{2}$

$= \cos n\theta + i\sin n\theta$

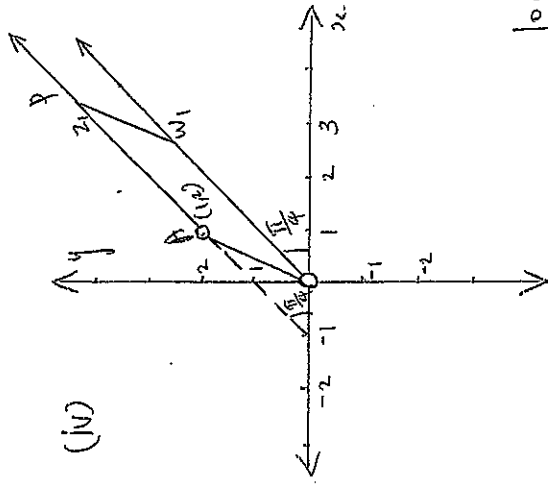
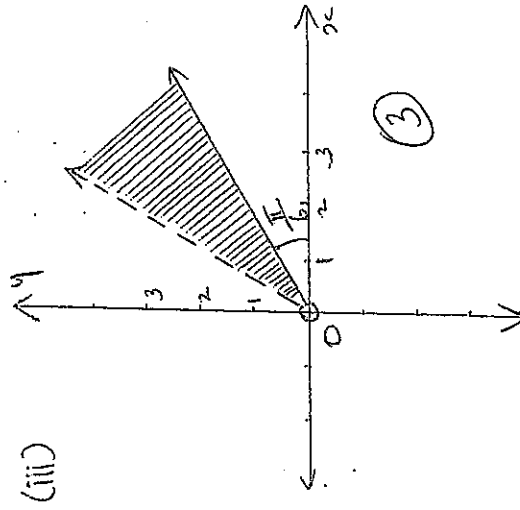
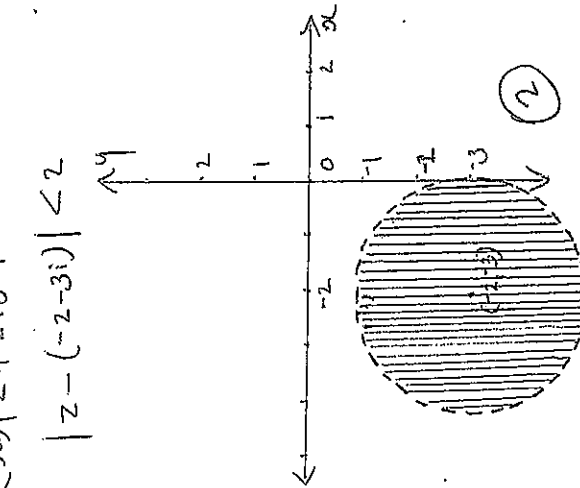
$= \text{RHS}.$

Question 7 (26 marks)



(b)(i) $|z + 2 + 3i| < 2$

$|z - (-2 - 3i)| < 2$



$\arg(z - (1 + 2i)) = \frac{\pi}{4}$

Gradient of AP = $\tan \frac{\pi}{4} = 1$

Equation of AP

$y - 2 = 1(x - 1)$

$y - 2 = x - 1$

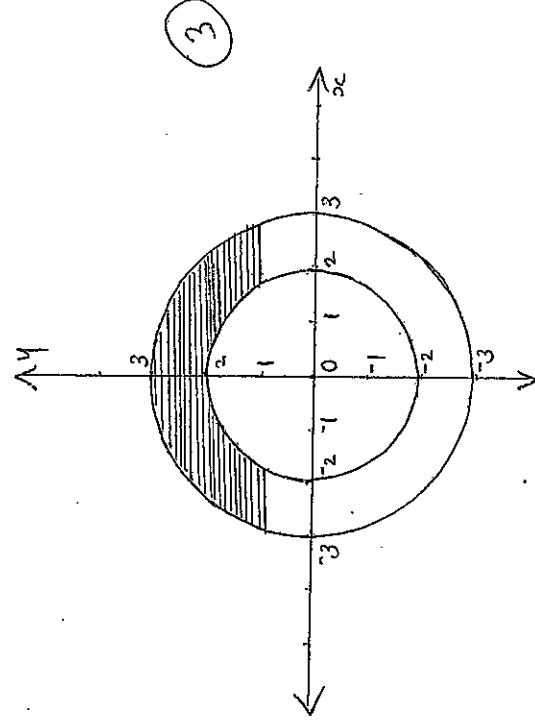
$y = x - 1 + 2$

$y = x + 1$

locus is the ray given by

$y = x + 1, x > 1, y > 2$

(v) $2 \leq |z| \leq 3$ and $\arg(z) \geq \frac{\pi}{4}$



$$(c) |z+w|^2 = (z+w)(\overline{z+w})$$

$$= (\overline{z}+\overline{w})(z+\overline{w})$$

$$= z\overline{z} + z\overline{w} + \overline{z}w + w\overline{w}$$

$$= |z|^2 + z\overline{w} + \overline{z}w + |w|^2 \quad \text{--- (1)}$$

$$|z-w|^2 = (\overline{z}-\overline{w})(z-w)$$

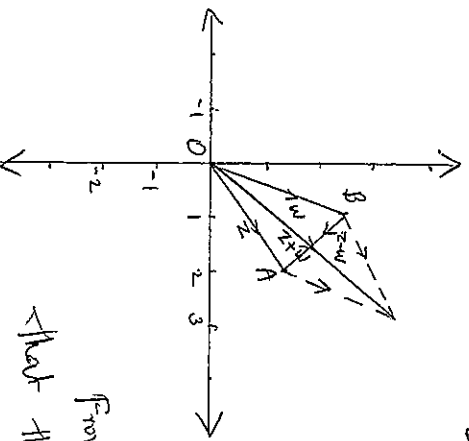
$$= (\overline{z}-\overline{w})(z-w)$$

$$= z\overline{z} - z\overline{w} - \overline{z}w + w\overline{w}$$

$$= |z|^2 - z\overline{w} - \overline{z}w + |w|^2 \quad \text{--- (2)}$$

Adding (1) and (2) we get

$$|z+w|^2 + |z-w|^2 = 2|z|^2 + 2|w|^2 = 2(|z|^2 + |w|^2)$$



$|z+w|$ and $|z-w|$ represent the lengths of the longer and shorter diagonals of the parallelogram.

From the above we conclude that the sum of the squares of the diagonals of a parallelogram is equal to the sum of the squares of the sides.

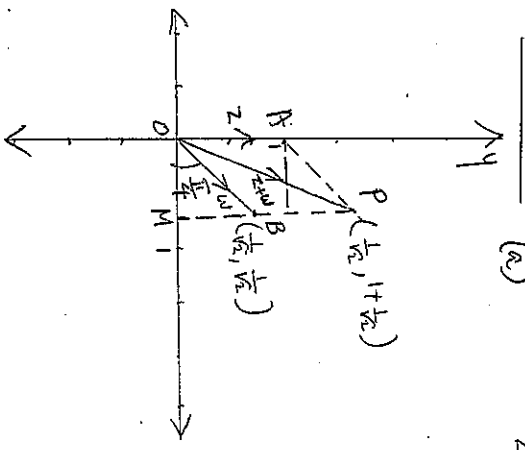
Question 8 (14 marks)

(a)

$\triangle AOP$ is isosceles ($AP = OA = 1$)

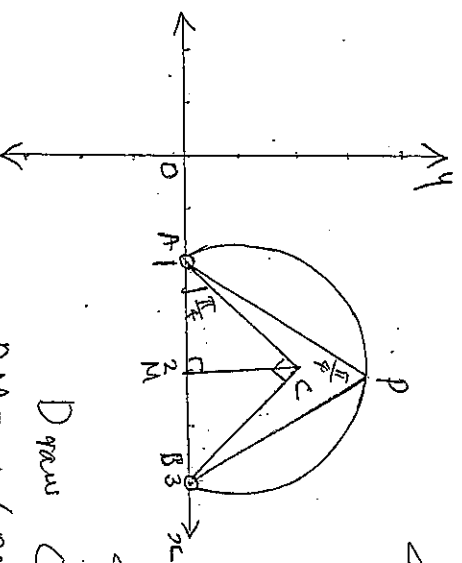
$$\angle AOP = \pi - \left(\frac{\pi}{2} + \frac{\pi}{4}\right) = \frac{\pi}{2}$$

$$\arg(z+w) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{3\pi}{4}$$



$$\tan \frac{3\pi}{4} = \frac{PM}{OM} = \frac{1 + \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} = \frac{\sqrt{2}+1}{1} = \sqrt{2}+1$$

$$(b) (i) \arg\left(\frac{z-3}{z-1}\right) = \frac{\pi}{4}$$



$\angle ACB = 90^\circ$ (angle at the centre is twice angle at the circumference).
 $\triangle ACB$ is isosceles ($AC = BC = \text{radius}$)
 $\angle CAB = \frac{180-90}{2} = 45^\circ$
 Draw $CM \perp OB$
 $AM = 1$ (perpendicular from the centre of a circle to a chord bisects the chord)

from 4.5 = $\frac{CM}{AM} = \frac{CM}{1}$

$1 = \frac{CM}{1} \therefore CM = 1$

Centre = (2, 1)

$AC = \sqrt{AM^2 + MC^2}$ (Pythagoras' theorem)

$= \sqrt{1^2 + 1^2} = \sqrt{2}$

radius = $\sqrt{2}$

Locus is the major arc APB of the circle with centre at (2, 1) and radius $\sqrt{2}$, excluding the points A and B.

(ii) $\left| \frac{z-1}{z+2} \right| = 2$

Let $z = x + iy$

$\left| \frac{z-1}{z+2} \right| = 2$ (4)

$|z-1| = 2|z+2|$

$|x+iy-1| = 2|x+iy+2|$

$|x-1+iy|^2 = 4|x+2+iy|^2$

$(x-1)^2 + y^2 = 4[(x+2)^2 + y^2]$

$x^2 - 2x + 1 + y^2 = 4[x^2 + 4x + 4 + y^2]$

(6)

$x^2 - 2x + 1 + y^2 = 4x^2 + 16x + 16 + 4y^2$

$4x^2 + 16x + 16 + 4y^2 - x^2 + 2x - 1 - y^2 = 0$

$3x^2 + 3y^2 + 18x + 15 = 0$

$x^2 + y^2 + 6x + 5 = 0$

$x^2 + 6x + y^2 = -5$

$x^2 + 6x + 9 + y^2 = -5 + 9$

$(x+3)^2 + y^2 = 4$

Locus is the circle with centre (-3, 0) and radius 2

Question 9 (14 marks)

(i) $z = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$

$z^5 = \cos 2\pi + i \sin 2\pi = 1$

$z^5 - 1 = 0$ (3)

$(z-1)(z^4 + z^3 + z^2 + z + 1) = 0$

Since $z \neq 1$, we have

$z^4 + z^3 + z^2 + z + 1 = 0$

(ii) $b^2 + b - 1$

$= \left(x + \frac{1}{x}\right)^2 + \left(x + \frac{1}{x}\right) - 1$

$= x^2 + 2 + \frac{1}{x^2} + x + \frac{1}{x} - 1$

$= x^2 + \frac{1}{x^2} + x + \frac{1}{x} + 1$ (2)

$= \frac{x^4 + 1 + x^3 + x + x^2}{x^2}$

$= 0$ (from (i))

(iii) $x^2 - bx + 1$

$= x^2 - \left(x + \frac{1}{x}\right)x + 1$

$= x^2 - x^2 - 1 + 1$

$= 0$

$x = \frac{b \pm \sqrt{b^2 - 4}}{2}$

$= \frac{b \pm \sqrt{-(4-b^2)}}{2}$

$= \frac{b \pm i\sqrt{4-b^2}}{2}$

$= \frac{b + i\sqrt{4-b^2}}{2}$

$(i = b - i\sqrt{4-b^2})$ is rejected because the imaginary part of x is positive

(iv) $b^2 + b - 1 = 0$ from (ii)

$b = \frac{-1 \pm \sqrt{1+4 \times 1}}{2} = \frac{-1 \pm \sqrt{5}}{2}$

$= \frac{-1 + \sqrt{5}}{2}$ ($\because \frac{-1 - \sqrt{5}}{2} < 0$ not possible)

We have

$x = \frac{b + i\sqrt{4-b^2}}{2}$

$\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} = \frac{b}{2} + \frac{i\sqrt{4-b^2}}{2}$

$\cos \frac{2\pi}{5} = \frac{b}{2} = \frac{-1 + \sqrt{5}}{4}$

(5)

$\sin \frac{2\pi}{5} = \frac{1}{2} \sqrt{4-b^2}$

$4-b^2 = 4 - (\sqrt{5}-1)^2 = 4 - \frac{(5+1-2)}{4}$

$= 4 - \frac{(6-2\sqrt{5})}{4} = 4 - \frac{(3-\sqrt{5})}{2}$

$= \frac{8-3+\sqrt{5}}{2} = \frac{5+\sqrt{5}}{2}$

$\therefore \sin \frac{2\pi}{5} = \frac{1}{2} \sqrt{\frac{5+\sqrt{5}}{2}}$